

Manuscript outline, checked against the draft of 2026-10-10

enceladus_DSM studies/keff_sintering/manuscript_outline.tex

2026-10-10

Framing. Sintering of ice by vapor transport, and the bulk conductivity that results, under conditions a laboratory can reach; Enceladus is the motivation and the closing implication. Every claim stays inside the simulated range: -40 to -5°C , air at 1 atm, sealed pores, $50\ \mu\text{m}$ grains, 30 days, two dimensions.

How to read this. Read against Literature/BaglinoJackson_DSM_manuscript.pdf (52 pages). **[in draft]** the draft already covers it; **[missing]** not there yet; **[revise]** there, but it no longer matches the model or the results. Boxes mark where a figure or table goes. Themes are prompts, not text.

Where the draft stands

Section	State
Key points	empty
Abstract, plain-language summary	marked “needs to be updated”; both still claim temperature-dependent kinetics, “without retuning” and a collapse across porosity
1 Introduction	complete in length; order and two paragraphs need work
2.1 Phase-field model	nearly complete
2.2 Gibbs–Thomson	thin: two equations; the clock, the thin-interface term and the α_c values are absent
2.3 Homogenization	complete
Numerical method; set-up of the aggregates	no section
3.1 Two-grain sintering	Molaro part written; saturated-pair paragraph is a placeholder
3.2 Aggregates, to Figure 6	written; the quantitative statements are absent
3.2 Aggregates, Figure 7	weakest part: stale text that contradicts the result
4 Discussion, 5 Conclusion	not yet reworked; 4.1 rests on a premise the model no longer has

Most in need of work, in order: (1) Section 3.2 from Figure 7 on; (2) Section 2.2; (3) the two missing Methods subsections; (4) the saturated-pair paragraph of 3.1; (5) the order of the Introduction.

Front matter

- **[revise] Title.** Still “Dry Snow Metamorphism ... Version 2”. The paper is no longer framed around snow.
- **[missing] Key points** (three). (i) For a given packing, temperature only rescales time. (ii) Conductivity follows specific surface area by one power law in well-connected packings; porosity sets the level. (iii) The model is the vapor route alone; what it does and does not reproduce of laboratory neck growth.
- **[revise] Abstract; Plain Language Summary.** Written last. Remove: kinetics “measured in experiments”; “without retuning”; “strong agreement” at -5°C ; collapse “regardless of ... initial porosity”.

1 Introduction

The draft’s eight paragraphs, and where each goes.

Draft	Content	Suggestion
D1	ice across the Solar System; what DSM is	keep as the opening, shortened
D2	snowpacks, avalanches, firn	cut to two sentences inside D1; it argues for a snow paper
D3	Enceladus, plume, Tiger Stripes	keep, move after the laboratory paragraphs (P5 below)
D4	sintering mechanisms, Figure 1, Molaro, Choukroun	split: mechanisms (P2) and laboratory evidence (P3)
D5	attachment kinetics, critical supersaturation, Libbrecht	reduce to two sentences in P4; the model uses a constant α_c , so the build-up to $\sigma_0(T)$ promises something the paper does not do
D6	DSM models: Kaempfer & Plapp, Bouvet, Braun	keep as P4
D7	gaps	rewrite: the gap is the link from microstructure to bulk property, not temperature-dependent kinetics
D8	this work	rewrite to the three results; it ends in a stray sentence fragment and says “28-day”

Target order.

- **P1. The problem** (from D1, D2). Ice aggregates change their thermal and mechanical properties as grains sinter.
- **P2. Mechanisms** (from D4). Sintering without densification; vapor transport (sublimation and desublimation) and surface diffusion. **[missing]** Say here that the model carries the first only. **[missing]** Define “desublimation”; the draft says vapor “deposits on the neck”.

Figure 1 **[in draft]** placed. Caption still says “neck of radius $R(t)$ ”; the figure now shows $w(t)$.

- **P3. What laboratory work has shown** (from D4). Molaro et al. (2019); Choukroun et al. (2020). **[missing]** Demmenie et al. (2025) is not in the Introduction.
- **P4. Models** (from D6, with two sentences of D5). Phase-field and sharp-interface models; each sets α_c against one data set.
- **[missing] P5. Why not natural snow.** Angular crystals at far higher porosity; overburden and densification. The analogues are unloaded ice-sphere and ice-powder samples.
- **P6. Motivation** (D3). Plume deposits on Enceladus.
- **P7. The gap and this work** (D7, D8). **[missing]** the sintering-age idea; **[missing]** the three results; **[revise]** “predictor ... dependent of the mean temperature or the initial porosity”.

2 Methods

2.1 Phase-field model of sintering by vapor transport

- **[in draft]** Fields, Eqs. (1a)–(1c), ρ_{vs}^I , material laws.
- **[missing]** What is *not* in the model: surface diffusion, densification.
- **[missing]** M_{sub} and α_{sub} in terms of τ_{sub} , λ and ε ; the temporal scaling ξ_T , ξ_v that the tables list but the equations do not show.
- **[revise]** Small things: “ $T_0 = 273.15^\circ\text{C}$ ” and “ $T_0 = 271.15\text{ K}$ ” (both 273.15 K); “Allen–Kahn”; heading still “Dry Snow Metamorphism”.

Table S1 \TableMaterialProperties. The draft has an *older* copy in the main text as its Table 1: it lacks the $\Delta\beta$ row and has $\tau_{\text{sub}} = \varepsilon^2\beta_{\text{sub}}/d_0$; its note points to two tables that are not in the draft (“Tables ?? and ??”). Re-import from `tables.tex`.

2.2 Interface kinetics and the Gibbs–Thomson condition

- **[in draft]** Eq. (6), the condition itself.
- **[revise]** **Eq. (7) has α_c in the numerator**; $\beta_{\text{sub}} \propto 1/\alpha_c$. Symbols differ from the tables (d_{sub} for d_0 , m_{mol} for m).
- **[missing]** The values: $\alpha_c = 1.020 \times 10^{-3}$ (aggregates), 0.1 (Molaro pairs), 1.002×10^{-3} (saturated pair), each with its β_{sub} .
- **[missing]** The thin-interface term $\Delta\beta$ and $\tau_{\text{sub}} = \varepsilon^2(\beta_{\text{sub}} + \Delta\beta)/d_0$, as in Kaempfer & Plapp (2009) and Moure & Fu (2024); what it costs (realised coefficient $\beta_{\text{sub}} + \Delta\beta$).
- **[missing]** The sintering age $\theta = t/\tau_{\text{sub}}$, defined here and not in passing in the Results; that ε is fixed across the study. *Open: state the clock as τ_{sub} at fixed ε , or as $\tau_R = \bar{R}^2\beta_{\text{sub}}/d_0$.*

2.3 Homogenization for the effective thermal conductivity

- **[in draft]** Cell problem, weak form, indicator function, tensor law (16).

- **[revise]** One sentence breaks off (“commonly referred to as co”); “tensorial”.

Figure 2 **[revise]** Present, but it floats into Section 3. Move after Eq. (17). Caption says contours at 0.05 and 0.95 and “red” lines; the figure draws 0.01, 0.5 and 0.99 in orange. Add that only the x problem is shown.

2.4 Numerical implementation

- **[missing]** The whole subsection: discretization (quadratic C^1 splines), time stepping and the step cap, sampling of $\mathbf{K}_{\text{eff}}$. Cite **Table S2** (\TableKeffParametersFull).
- **[missing]** Checks, one sentence each with a pointer to the supplement: domain size (Fig. S1), interface width (Fig. S2), time step and sampling (Fig. S3).

2.5 Aggregates: packings, boundary conditions, simulation matrix

- **[revise]** Now two sentences inside Section 3.2. Move here and expand.
- **[in draft]** Drop-and-roll deposition (named). **[missing]** void filling; five packings per porosity; acceptance tests.
- **[missing]** Why periodic boundaries: scale separation, no wall physics to invent, a closed system driven by curvature alone.
- **[missing]** The porosities and temperatures as lists; the master simulation used for single-run figures.

3 Results

[revise] Opening paragraph. Promises conductivity “independent of the mean temperature as well as the initial porosity” and a model that “captures sintering dynamics for varying temperatures”. Restate as the three results.

3.1 Two-grain sintering

Unsaturated pair (Molaro et al., 2019).

- **[in draft]** Why this experiment; set-up; humidity inferred from the grain shrinkage; α_c at the upper end of the range; the authors’ caveat on the -5°C data.
- **[missing]** The numbers: 41% of the measured neck growth at -20°C ; 170% at -5°C (52.1 against $44.0\ \mu\text{m}$).
- **[missing]** That the -20°C shrinkage is a calibration, so panels (c) and (d) are not a test at that temperature; neither the text nor the caption mentions those panels.
- **[revise]** “(cite other works as well)”;

Table S3 \TableGrainPairParameters, cited where the draft has “table ??”.

Figure 3 **[in draft]** placed. Caption stops at (b): add (c), (d).

Saturated pair (conditions of Demmenie et al., 2025).

- **[missing]** The paragraph (the draft has “Insert paragraph”). Content: equal grains, vapor saturated over the grains’ own curvature; grain size holds (+0.06%); exponent 0.21 after relaxation (0.21–0.23 by window) against 0.26–0.33 measured and 1/3 in theory.
- *Open: interpretation. The Gibbs–Thomson prediction for this run is 0.29 and matches the final width to 4%; the run is faster early, where the neck fillet is about two interface widths across. Vapor transport alone gives 0.25–0.33 at Demmenie’s own conditions for $\alpha_c \leq 0.1$. Parallel mechanisms add rates, so missing surface diffusion cannot raise an exponent.*

Figure 4 **[in draft]** placed, with caption.

3.2 Aggregates

Set-up.

- **[in draft]** 125 runs, periodic, 2 mm, 30 days. **[revise]** mesh “ 2829×2928 ” (2829 both ways).

Table 1 **[missing]** \TableKeffParameters. After the sentence that states the matrix.

What sintering does to the aggregate.

- **[in draft]** Description of the gallery.

Figure 5 **[in draft]** placed, with caption.

Temperature sets the clock.

- **[in draft]** Panels (b)–(d) described in words.
- **[revise]** Off-diagonal terms “at least one order of magnitude lower”: k_{xy} reaches about 12% of k_{xx} on one packing; zero in the mean.
- **[missing]** The result stated: for every packing, k_{eff} at matched SSA agrees across the five temperatures within 0.27% (Fig. S4); rate follows τ_{sub} within 4%.
- **[revise]** “the trend is linear”: linear in $\log \theta$, about 0.13 per decade, over a stated range.

Table 2 **[missing]** \TableKeffTemperature. After θ is introduced, before Figure 6.

Figure 6 **[in draft]** placed. **[revise]** Caption says “normalized by its initial value” (now by the value at $\theta_r = 30$) and $|\sigma| \leq 0.31$ (the bar reads 0.34); define θ_r .

Conductivity as a function of the microstructural state.

- **[revise]** “3 packings for each porosity”: five packings for each of three porosities.
- **[revise]** High porosities excluded because the ice “remains disconnected in the horizontal direction”: not shown; what is shown is a weaker, scattered exponent and k_{xx}/k_{yy} of 0.79 and 0.64.
- **[revise]** Final paragraph (“28 days”, “in ??”, “no longer is dependent of porosity”) contradicts the figure: porosity sets the level; only the exponent is shared.
- **[missing]** The numbers: exponent -0.80 (per porosity $-0.79, -0.82, -0.78$); level $1.370 \exp(-5.01 \varphi)$ with 6.6% scatter; a factor of about 3.3 in k_{eff} across porosity at 30 d.

- **[missing]** The rise table by porosity and temperature; the two densest packings tie.

Figure 7 **[in draft]** placed. **[revise]** Caption for (a) says a single seed with seed scatter (it is the mean of five with their range) and stops at “(c)”.

4 Discussion

State of the draft. Two subsections, written for the earlier model. Section 4.1 argues that $\beta_{\text{sub}}(T)$ comes from Libbrecht’s $\sigma_0(T)$, that nothing was retuned, and that -5°C is “a genuine prediction”. With a constant α_c none of that holds; the section has to be rebuilt, not edited. Section 4.2 claims a collapse across porosity, isotropy, and sintering “on the order of decades” near the Tiger Stripes. About 40 citations print as “?”.

4.1 Temperature dependence of the kinetics

- Why the collapse holds: attachment-limited transfer, fixed α_c , one discretization; what would break it.
- Activation energy of the vapor route, about 50 kJ mol^{-1} , against 24 measured by Choukroun et al.; the ranges overlap at 233–243 K.
- Reusable from the draft: the survey of how others set α_c (Bouvet, Braun) and the point that it is an effective parameter.

4.2 SSA as the state variable

- What the power law means; why porosity alone is not enough.
- Loss of the law at high porosity; anisotropy as a consistency check.
- Reusable from the draft: the comparison with Calonne et al. (density alone is incomplete; conductivity changes at constant density).

4.3 What the vapor route captures and what it lacks

- Figures 3 and 4 read together. Surface diffusion and the quasi-liquid layer; rate versus exponent.

4.4 What a laboratory could measure

- Time–temperature superposition and its activation energy; k_{eff} against SSA on unloaded ice-sphere samples; SSA against log time. Reusable from the draft: how SSA is measured.

4.5 Implications for plume deposits on Enceladus

- Qualitative only; no timescales in years. Reusable from the draft: the Choukroun and radar-attenuation paragraphs, without the “decades” claim.

4.6 Limitations

- Two dimensions; air-filled pores; one grain size; vapor route only; first day or so depends on the interface width; porosity ≤ 0.375 . Reusable from the draft: conduction only; constant K_{ice} , K_{air} .

5 Conclusions

- **[revise]** Not yet reworked. Three results, each with its range of validity; the most useful experiment to do next.

Open Research; Acknowledgments

[missing] Code and data availability; run inventory (`RUN_TABLE.md`).

Supporting Information

- **Tables.** S1 material properties; S2 complete simulation parameters; S3 two-grain parameters.
- **Figures.** S1 domain-size convergence; S2 interface width; S3 time step and sampling; S4 temperature collapse for all packings; S5 porosity trends, SSA by porosity, anisotropy in time.
- **Text.** Packing construction; run table.